

Variance of a Random Variable

For a random variable X with mean $E(X) = \mu$, the variance of X is defined as the expected value of $(X - \mu)^2$. That is,

$$\text{Var}(X) = E[(X - \mu)^2]$$

- Variance is the average squared deviation of X from its mean
- Measure of variability. The larger the variance, the more spread out around the mean the values the random variable can take on

Proof

$$\begin{aligned} E[(X - \mu)^2] &= E[X^2 - 2\mu X + \mu^2] \\ &= E(X^2) - 2\mu E(X) + \mu^2 \\ &= E(X^2) - 2\mu^2 + \mu^2 \\ &= E(X^2) - \mu^2 \end{aligned}$$

Note

$E(c) = c$ for any constant c

Standard Deviation of a Random Variable

- Positive square root of $\text{Var}(X)$ for a random variable X
- Convention denotes variance as $\text{Var}(X) = \sigma^2$. Therefore, standard deviation is often denoted by σ
- Scale of variance is square of that of the mean. In comparison, the standard deviation is in the same units as the mean

Example

Consider two users. One receives either 8 or 12 emails per day, with a 50-50 chance. The other receives 0 or 20 emails, also with a 50-50 chance.

Find the expectation, variance, and standard deviation of these two random variables

$$X \begin{cases} E(X) = 8 \cdot \frac{1}{2} + 12 \cdot \frac{1}{2} = 10, E(X^2) = 8^2 \cdot \frac{1}{2} + 12^2 \cdot \frac{1}{2} = 104 \\ \text{Var}(X) = \sigma^2 = E(X^2) - E(X)^2 = 104 - 100 = 4 \\ \text{Standard Deviation} = \sqrt{\text{Var}(X)} = \sqrt{4} = 2 \end{cases}$$

$$Y \begin{cases} E(Y) = 20 \cdot \frac{1}{2} + 0 \cdot \frac{1}{2} = 10, E(Y^2) = 20^2 \cdot \frac{1}{2} + 0^2 \cdot \frac{1}{2} = 200 \\ \text{Var}(Y) = \sigma^2 = E(Y^2) - E(Y)^2 = 200 - 100 = 100 \\ \text{Standard Deviation} = \sqrt{\text{Var}(Y)} = \sqrt{100} = 10 \end{cases}$$

Theorem

Let X be a random variable with finite mean μ and variance σ^2 . Then, for all constants a and b ,

$$\text{Var}(aX + b) = a^2 \text{Var}(X)$$

$$E[(X - \mu)^2]$$

Proof

$$\begin{aligned} \text{Var}(aX + b) &= E[(aX + b - E(aX + b))^2] \\ &= E[(aX + b - (aE(X) + b))^2] \\ &= E[(a(X - E(X)))^2] \\ &= E[a^2(X - E(X))^2] \\ &= a^2 E[(X - E(X))^2] \\ &= a^2 \text{Var}(X) \end{aligned}$$

Moment Generating Functions

Let X be a random variable such that for some $h > 0$, $E(e^{tx})$ exists for $-h < t < h$. The moment generating function (mgf) of X is defined by the function

$$M(t) = E(e^{tx}), \text{ for } -h < t < h$$

- Unique to each random variable
- Can uniquely identify distributions. This means that if X and Y have the same distribution ($F_X(z) = F_Y(z)$ for all z). Then, $M_X(t) = M_Y(t)$ in a neighborhood of 0. The converse is also true
- If we can find $E(e^{tx})$, we can find $E(X^k)$. We call $E(X^k)$ the k^{th} moment of the distribution, or the k^{th} moment of X
- Proves that a random variable possesses a particular probability distribution $p(x)$. If $M(t)$ exists for a probability distribution $p(x)$, it is unique

Example

Let $M(t) = \frac{1}{6}e^t + \frac{2}{6}e^{2t} + \frac{3}{6}e^{3t}$. Find $E(X)$, $\text{Var}(X)$, and the distribution of X

$$E(X) = \frac{d}{dt} \left[\frac{1}{6}e^t + \frac{2}{6}e^{2t} + \frac{3}{6}e^{3t} \right]_{t=0} = \left[\frac{1}{6}e^t + \frac{4}{6}e^{2t} + \frac{9}{6}e^{3t} \right]_{t=0} = \frac{14}{6}$$

$$E(X^2) = \frac{d^2}{dt^2} \left[\frac{1}{6}e^t + \frac{2}{6}e^{2t} + \frac{3}{6}e^{3t} \right]_{t=0} = \left[\frac{1}{6}e^t + \frac{8}{6}e^{2t} + \frac{27}{6}e^{3t} \right]_{t=0} = 6$$

$$\text{Var}(X) = E(X^2) - \mu^2 = 6 - \left(\frac{14}{6}\right)^2 \approx 0.556$$

$$E(e^{tx}) = \sum e^{tx} p(x) = \frac{1}{6}e^t + \frac{2}{6}e^{2t} + \frac{3}{6}e^{3t}$$

x	1	2	3
$p(x)$	$\frac{1}{6}$	$\frac{2}{3}$	$\frac{3}{6}$

* In general, we can't find pmf/pdf from mgf

Theorem

If $M(t)$ exists, then for any positive integer k

$$\frac{d^k}{dt^k} [M(t)]_{t=0} = M^{(k)}(0) = E(X^k)$$

- If you find the k^{th} derivative of $M(t)$ with respect to t and set $t=0$, the result is $E(X^k)$